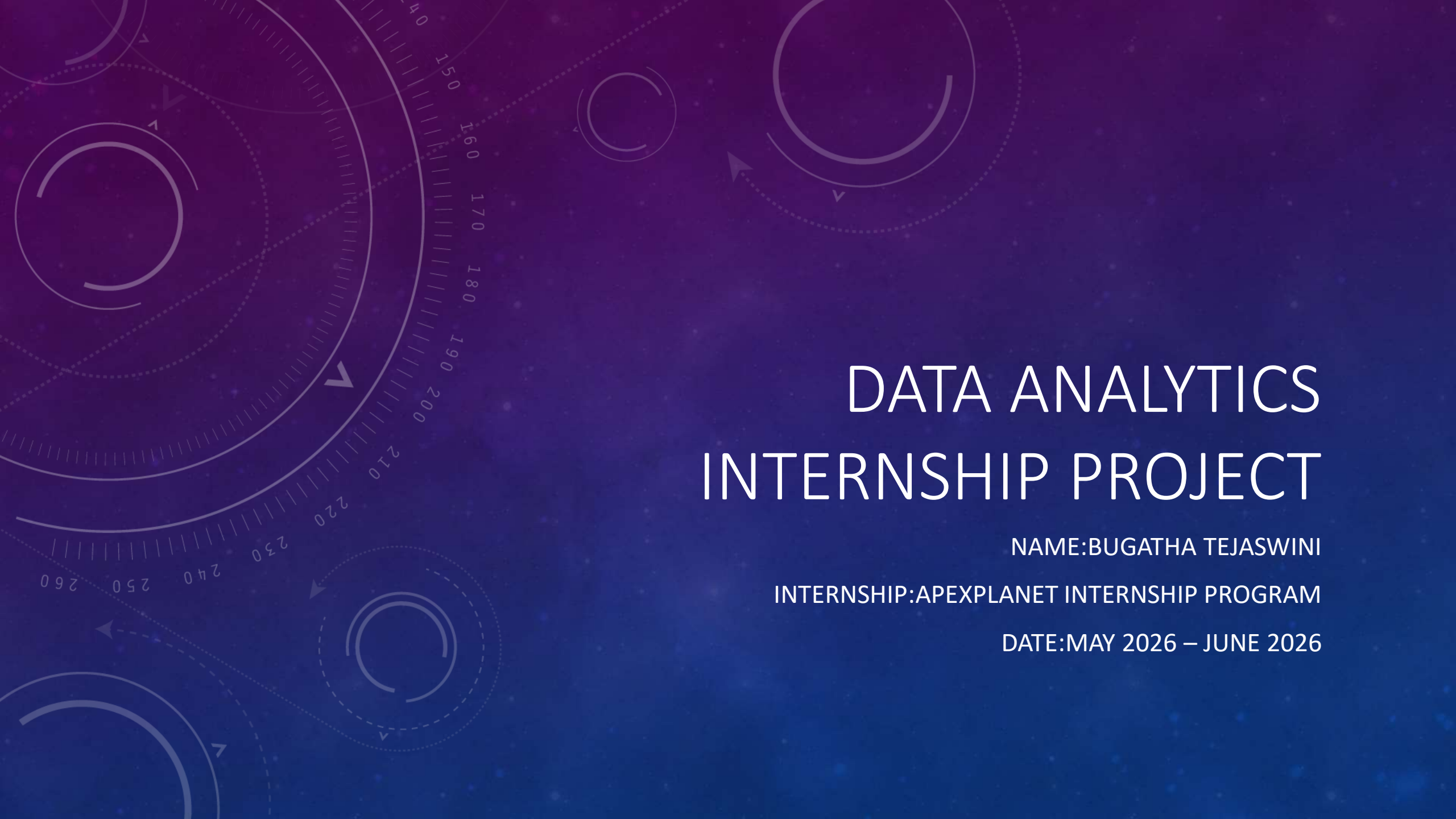




DATA ANALYTICS INTERNSHIP PROJECT

NAME:BUGATHA TEJASWINI

INTERNSHIP:APEXPLANET INTERNSHIP PROGRAM

DATE:MAY 2026 – JUNE 2026

PROJECT INTRODUCTION:

- This project focuses on analyzing the Superstore Sales dataset to identify business trends, sales performance, customer behavior, and regional profitability using data analytics tools and visualization techniques.
- The project includes data cleaning, exploratory data analysis, SQL querying, dashboard creation, and business storytelling.

OBJECTIVES:

- Analyze sales and profit trends
- Identify top-performing product categories
- Compare regional sales performance
- Understand customer segments
- Build interactive dashboards
- Generate business insights for decision-making

TOOLS USED:

Tool

Purpose

Python

Data Cleaning & EDA

MySQL Workbench

SQL Query Analysis

Excel

Dashboard Creation

Power BI

Interactive Dashboard

GitHub

Project Repository

DATASET OVERVIEW:

- **Dataset Contains:**

- Orders data
- Product categories
- Customer segments
- Regional sales information
- Profit and discount details

- **Important Columns:**

- Sales
- Profit
- Category
- Region
- Segment
- Order Date

DATA CLEANING PROCESS:

- **Content**
- **Steps Performed:**
 - Imported dataset using Python
 - Checked missing values
 - Removed duplicate records
 - Corrected inconsistent formats
 - Converted date columns
 - Prepared analysis-ready dataset
- **Mention:**
 - Libraries used:
 - Pandas
 - NumPy

- **Achievements:**
- Improved data quality
- Standardized dataset structure
- Reduced inconsistencies
- Enabled accurate analysis

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[illegible]

Global Supply Operations - Microsoft Excel

Summary: JAN 15 2016

Total Sales	Total Profit	Total Order	Avg. Discount
3,700,000	884,512.4	8077	0.134078

Insights

- Consumer segment generated highest sales
- West region showed strong profit performance
- Food and Office Supplies contributed major sales
- Quarterly offered overall profitability

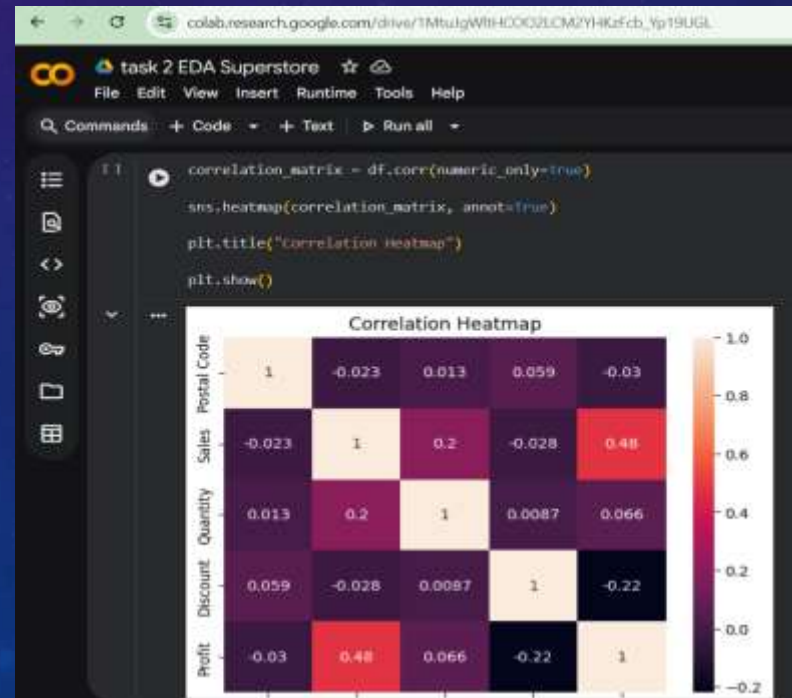
EXPLORATORY DATA ANALYSIS:

- **Analysis Performed:**
- Sales by Category
- Profit by Region
- Sales by Segment
- Yearly Sales Trend
- Discount Impact Analysis

```
1 Segment 9977 non-null object
2 Country 9977 non-null object
3 City 9977 non-null object
4 State 9977 non-null object
5 Postal Code 9977 non-null int64
6 Region 9977 non-null object
7 Category 9977 non-null object
8 Sub-Category 9977 non-null object
9 Sales 9977 non-null float64
10 Quantity 9977 non-null int64
11 Discount 9977 non-null float64
12 Profit 9977 non-null float64
dtypes: float64(3), int64(2), object(8)
memory usage: 1813.4+ KB
```

```
df.describe()
```

	Postal Code	Sales	Quantity	Discount	Profit
count	9977.000000	9977.000000	9977.000000	9977.000000	9977.000000
mean	55154.964117	230.148902	3.790719	0.156278	28.69013
std	32058.266816	623.721409	2.226657	0.206455	234.45784
min	1040.000000	0.444000	1.000000	0.000000	-5599.97800
25%	23223.000000	17.300000	2.000000	0.000000	1.72620
50%	55901.000000	54.816000	3.000000	0.200000	8.67100
75%	80008.000000	209.970000	5.000000	0.200000	29.37200



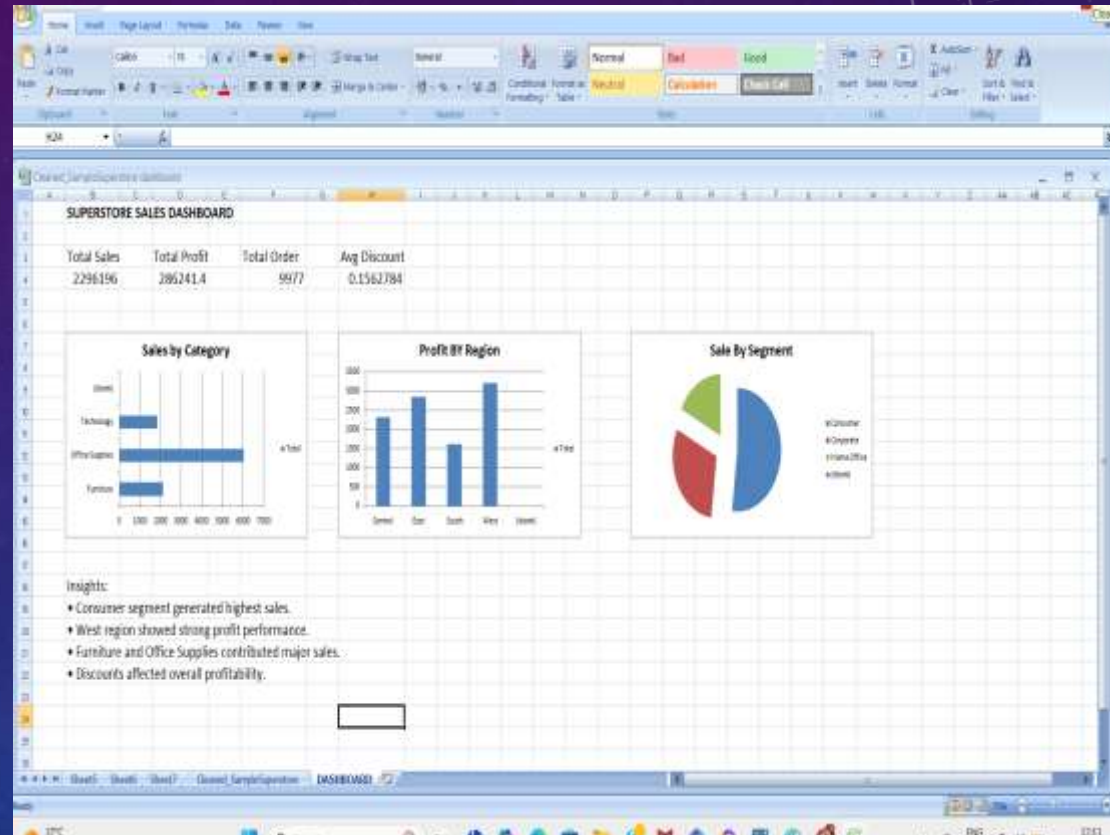
Excel Dashboard Analysis:

An interactive Excel dashboard was created to visualize sales performance, regional profit distribution, and customer segment contribution.

The dashboard provides a quick overview of important business metrics and helps identify sales trends and profitability patterns.

EXCEL DASHBOARD:

- **Dashboard Highlights**
- Sales by Category analysis
- Profit by Region comparison
- Sales contribution by Segment
- KPI summary metrics
- Business insights section



SQL Business Analysis:

SQL queries were used to extract meaningful business insights from the Superstore dataset. Different queries were written to analyze sales trends, category performance, and regional profitability.

Business Questions Solved

Which category generated highest sales?

Which region produced highest profit?

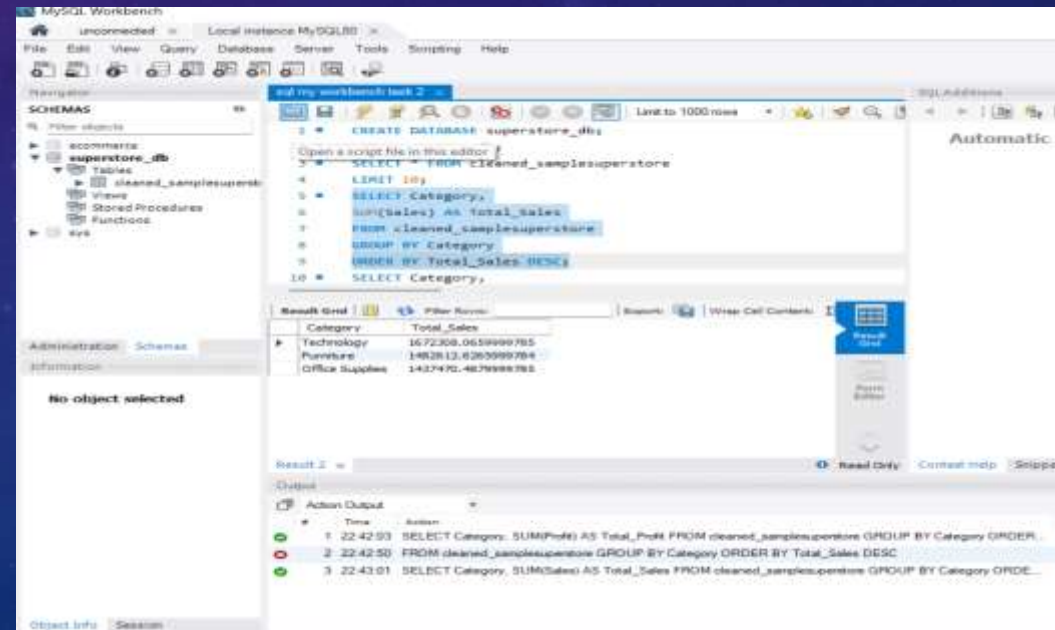
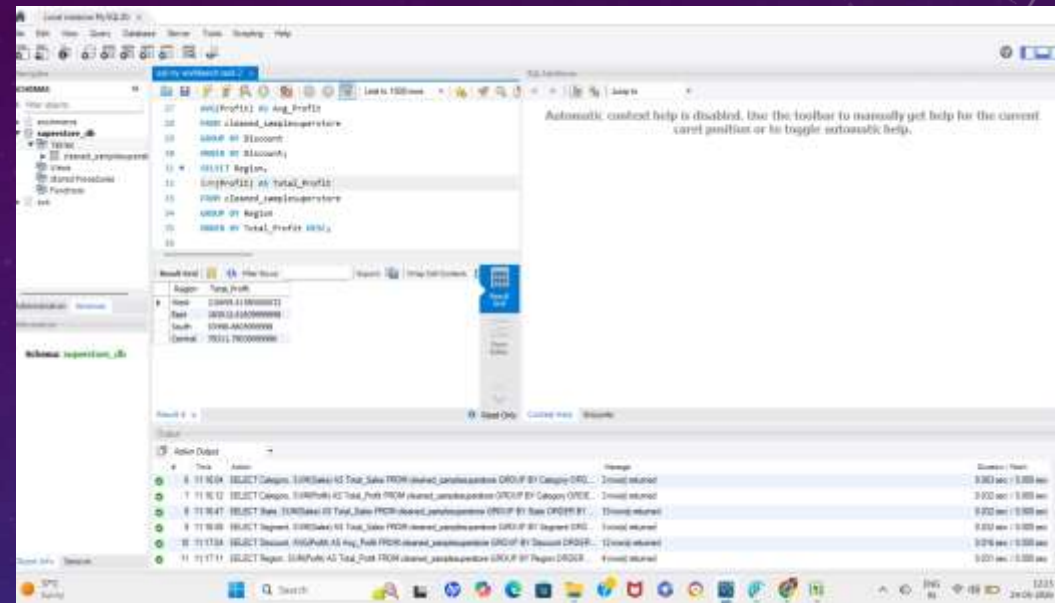
What is the yearly sales trend?

Which customer segment contributed most sales?

Which products generated maximum revenue?

SQL BUSINESS ANALYSIS:

- SQL Operations Used
- SELECT
- WHERE
- GROUP BY
- ORDER BY
- SUM()
- AVG()
- COUNT()



INTERACTIVE POWER BI DASHBOARD:

- **Features:**
- Dynamic filters
- KPI cards
- Region-wise analysis
- Category-wise analysis
- Segment filtering
- Interactive visuals



KEY PERFORMANCE INDICATORS:

KPI	Value
Total Sales	2.26 Million
Total Orders	10K
Best Performing Region	West
Top Segment	Consumer
Highest Sales Category	Technology

BUSINESS INSIGHTS:

- Consumer segment generated highest sales
- West region achieved strong sales performance
- Technology category contributed highest revenue
- Discounts impacted profitability
- Sales showed growth trend over years

BUSINESS STORY:

- The analysis reveals that the business performed strongly in the West region, especially in the Technology category. Consumer customers contributed major sales revenue, while discounts reduced overall profitability in some categories.
- Interactive dashboards helped identify important trends and business opportunities.

STATISTICAL VALIDATION:

- **Business Hypothesis:**
- West region sales are significantly higher than South region sales.
- **Null Hypothesis (H_0)**
- *West Sales = South Sales*
- **Alternative Hypothesis (H_1)**
- *West Sales > South Sales*
- If:
- $p < 0.05$
- then result is statistically significant.
- $p < 0.05$

STATISTICAL CONCLUSION:

- The statistical testing indicates that the West region performed significantly better in sales compared to the South region.
- This supports the business insight obtained from the dashboard analysis.

CHALLENGES & LEARNING:

- **Challenges:**

- Handling missing data
- Building meaningful visualizations
- Understanding business KPIs
- Dashboard formatting

- **Skills Learned:**

- Data Cleaning
- SQL Query Writing
- Dashboard Design
- Business Analytics
- Data Storytelling

CONCLUSION:

- This project successfully transformed raw sales data into actionable business insights using data analytics tools and visualization techniques.
- The dashboards and analysis helped understand customer behavior, sales trends, and regional performance effectively.

FUTURE SCOPE:

- **Content**
- Add predictive analytics
- Create real-time dashboards
- Perform customer segmentation
- Use machine learning models
- Build automated reporting systems

THANK YOU

- PROFILE LINKS:
- GITHUB LINK: <https://github.com/tejaswinibugatha>
- LINKEDIN LINK: https://www.linkedin.com/in/teju-bugatha-744a692b6?utm_source=share_via&utm_content=profile&utm_medium=member_android
- “THANK YOU FOR ATTENTION”